

Electromagnetic Induction Problem Generator

Faraday's Law Analysis: $\vec{B}(t) \rightarrow \Phi(t) \rightarrow e(t) \rightarrow i(t)$

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Coil Parameters

Turns (N)



Radius r (cm)



Resistance R (Ω)



Problem Statement

A circular coil with **10** turns and radius **10 cm** is placed in a uniform magnetic field $\vec{B}$. The total resistance of the circuit is **0.5 Ω** .

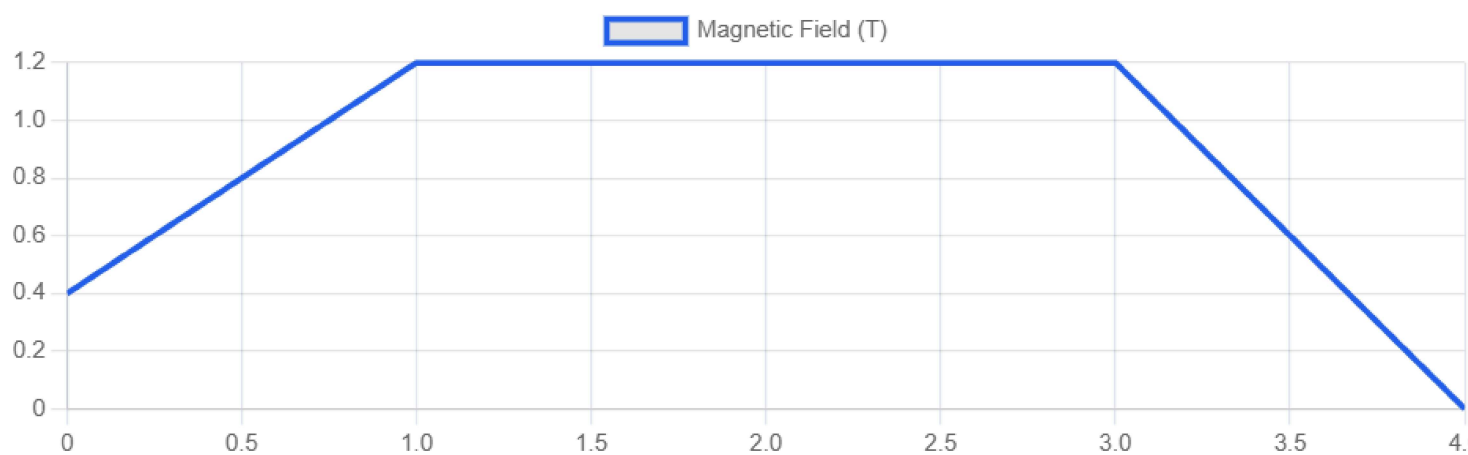
The vector $\vec{B}$ is perpendicular to the plane of the coil. Its magnitude varies with time as shown in the graph.

- $t = 0 \rightarrow 1\text{s}$: Linear increase (Rising Phase)
- $t = 1 \rightarrow 3\text{s}$: Constant Field (Plateau)
- $t = 3 \rightarrow 4\text{s}$: Linear decrease (Falling Phase)

Questions:

- Determine the direction of induced current in each phase (Lenz's Law).
- Calculate the Flux Φ at $t = 0.5\text{s}$ and $t = 3.5\text{s}$.
- Calculate the induced EMF (e) and current (i) for the rising and falling phases.

Magnetic Field B vs Time



Teacher Analysis Hidden

Enable Teacher Mode to see Flux, EMF, and Current graphs.